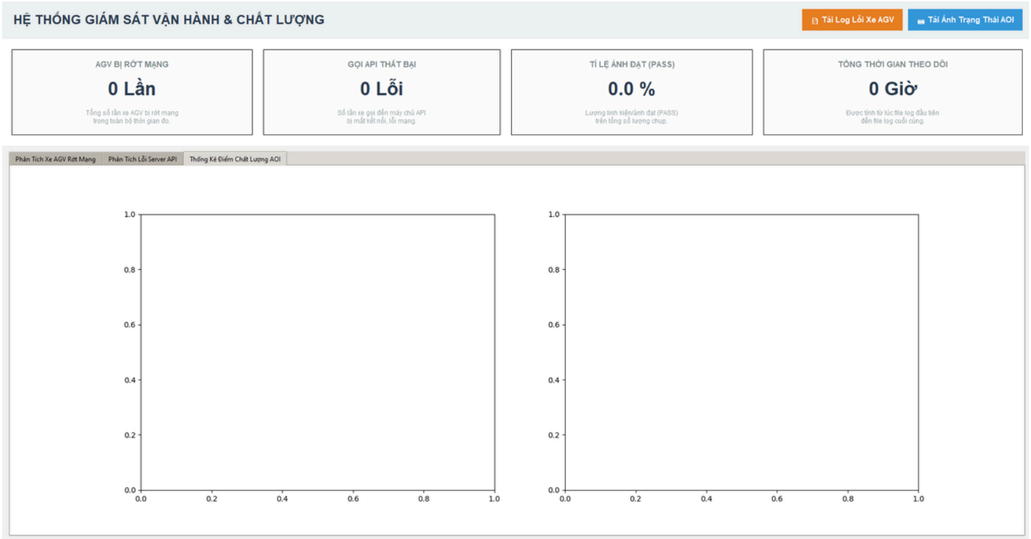
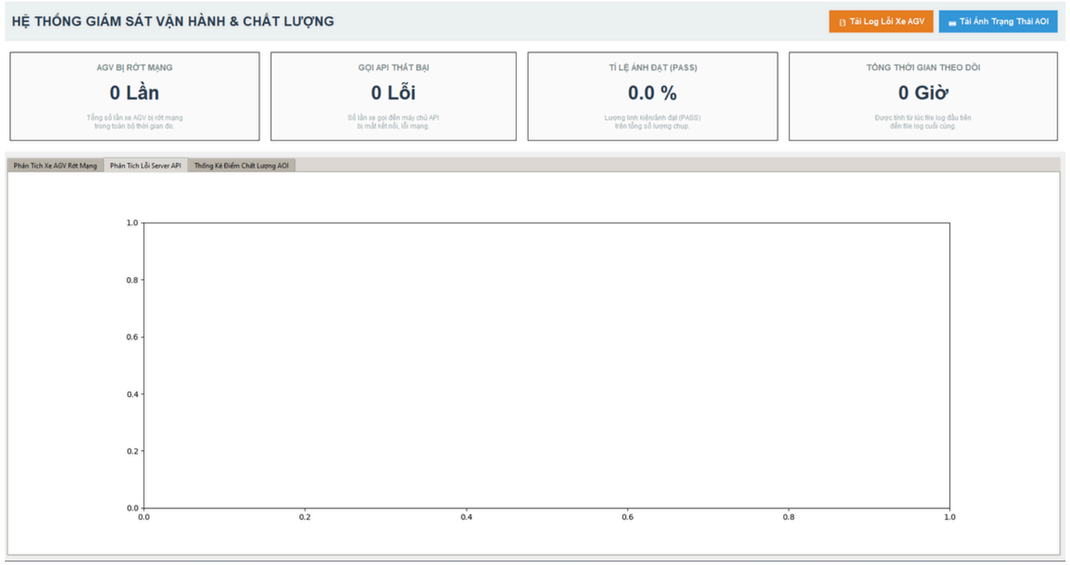
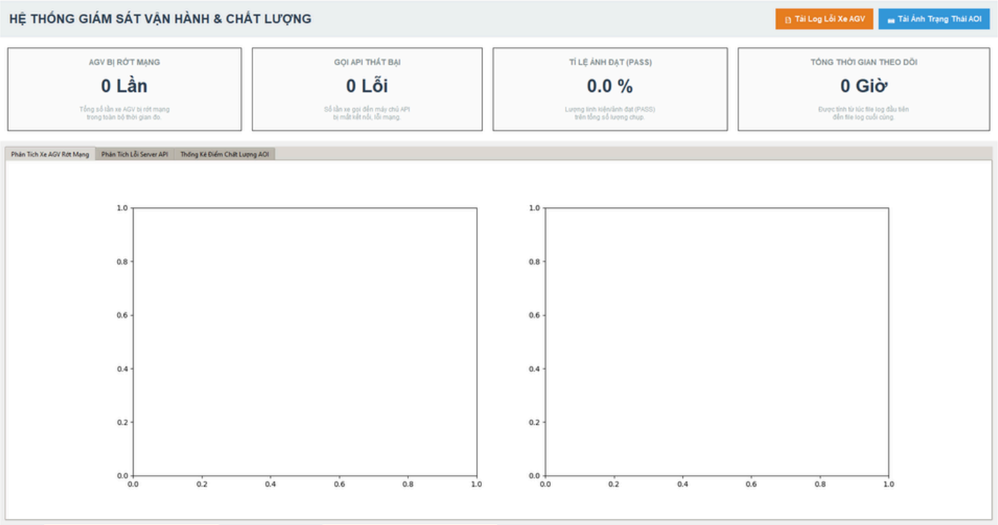
BG6：AOI 和 AGV 看板







Presented by:
Pham Minh Quang
范明光

Pegatron Corp

PROJECT OVERVIEW

- Objective: Smart Monitoring & Yield Analysis.
- Technologies: Python, Pandas, Matplotlib.

CHALLENGES

- Automated error detection in AGVs.
- Unified quality tracking via AOI images.
- Conversion of raw data into visual trends.

Modular Design

- Core Engine: Separates data analysis logic from the user interface (UI).
- Specialized Modules: **agv.py** handles vehicle logs; **aoi.py** processes images.
- Data Processing: Uses Pandas to process large datasets with high performance.

```
factory_dashboard/
 ├─── main.py #  Main UI & Dashboards
 ├─── agv.py # AGV Log Parsing & Metrics
 ├─── aoi.py # AOI Image Parsing & Metrics
```

Workflow

1. Input: Batch scanning (scanning multiple files at once).
2. Processing: Real-time timestamp parsing (extracting time data as it happens).
3. Display: Automatic UI updating (the interface refreshes itself automatically).

main.py

```python
def load_agv(self):
    files = filedialog.askopenfilenames(title="Chọn log AGV")
    if files:
        self.df_offline, self.df_api, _, self.coverage = agv.parse_agv_logs(files)
        self.update_kpi()
        self.draw_agv_charts()
```

Log Parsing

- Offline Detection: Uses Regular Expressions (Regex) to capture vehicle IDs.
- API Analysis: Filters failed connection requests from the source string.
- Incident Frequency: Generates error statistics based on production time slots.

agv.py

```python
offline_match = re.search(r'(\d+)号AGV已经掉线', line)
if offline_match:
    agv_id = offline_match.group(1)
    offline_events.append({'TS': timestamp, 'AGV': agv_id})


df_offline = _offline.groupby(['Hour', 'AGV']).size().reset_index(name='Count')
```

Yield Rate Analysis

- Classification: Identifies PASS/FAIL status from image filenames.
- Time Correlation: Links production batches to their corresponding dates.
- Rate Statistics: Calculates the exact percentage of parts that meet standards.

agv.py

```
status = 'PASS' if 'ALL PASS' in filename.upper() else 'FAIL' if 'FAIL' in filename.upper() else 'UNKNOWN'

summary = df.groupby(['Date', 'Status']).size().unstack(fill_value=0).reset_index()
```

(UX) & Visualization

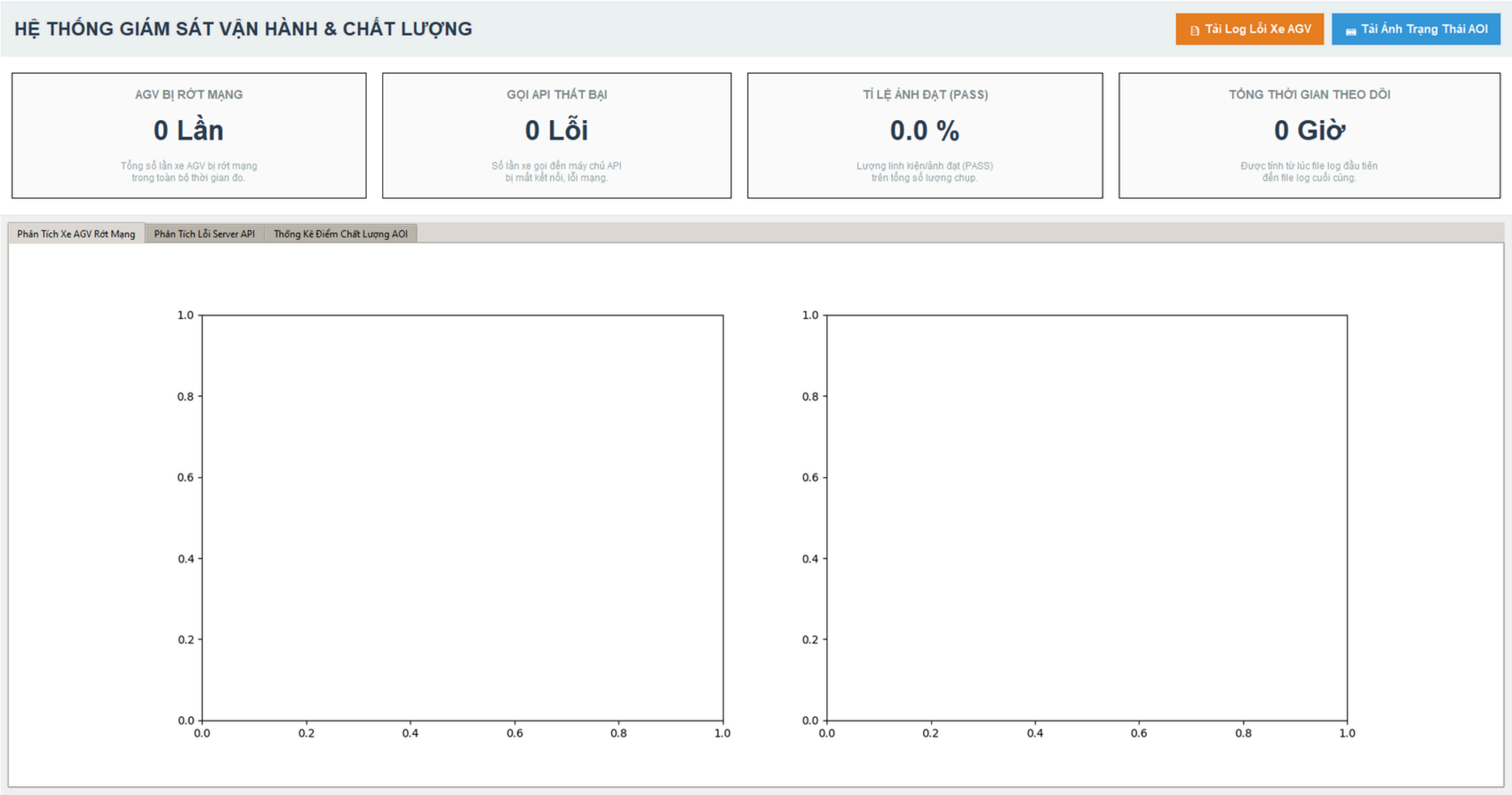
- KPI Alerts: Uses smart coloring (Green/Red) to signal system incidents.
- Matplotlib Charts: Renders graphs directly onto the control dashboard.
- Data Labels: Displays actual values directly on the charts.

agv.py

```python
# Cập nhật màu sắc cảnh báo linh hoạt trên các ô KPI
color = "#e74c3c" if total_off > 0 else "#27ae60" # Đỏ nếu có sự cố
self.kpis['agv']['val'].config(text=f"{total_off} Lần", fg=color)

# Tự động gắn nhãn (Label) lên từng cột biểu đồ bar chart
bars = self.ax_agv_top.bar(top['AGV'].astype(str), top['Count'])
self.ax_agv_top.bar_label(bars, padding=3, fontsize=9)
```

DASHBOARD OVERVIEW



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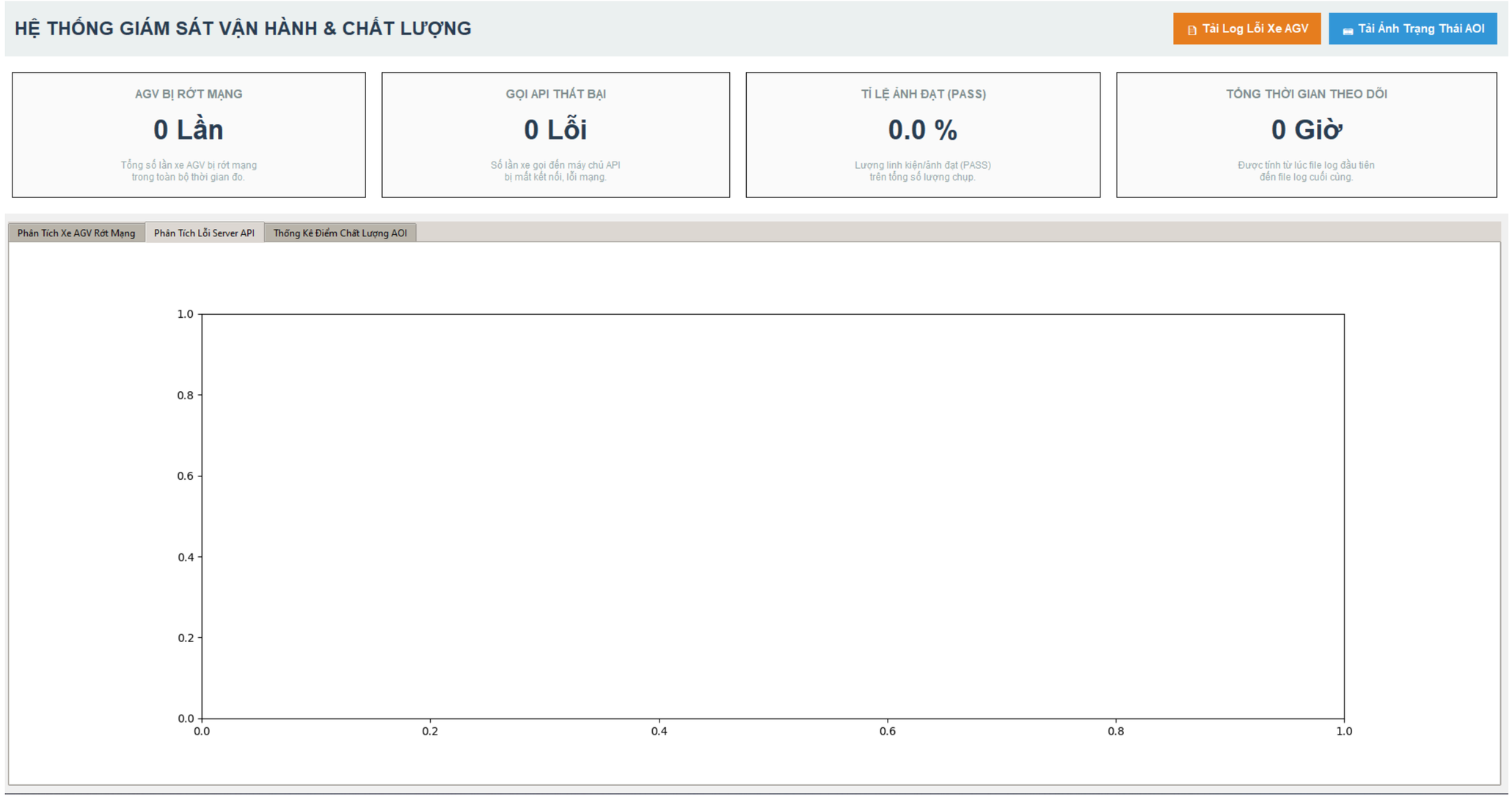

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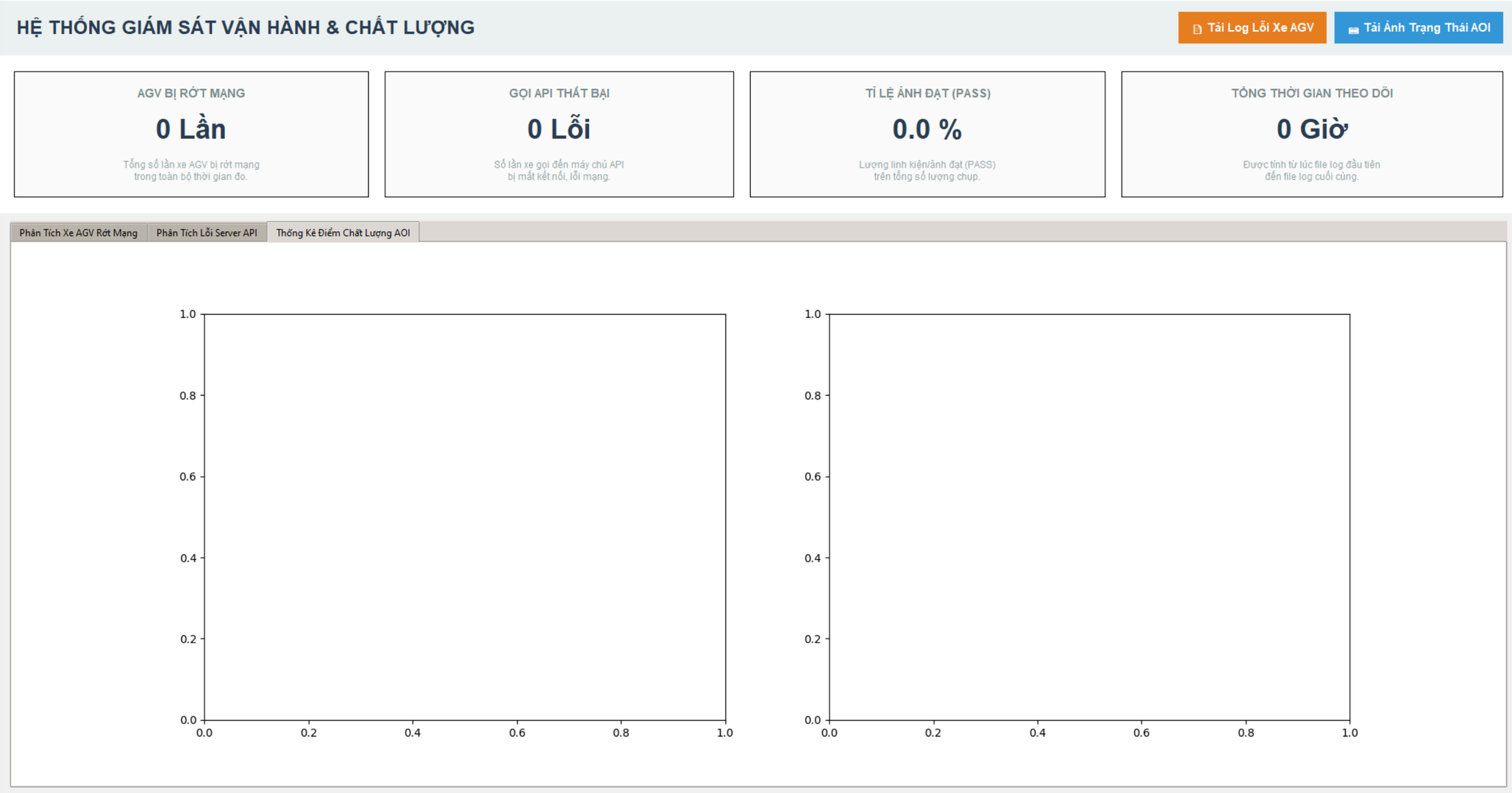

Current Limitations & Operational Status

- Access Restrictions: The internal database is under strict corporate control. We do not have the necessary permissions to establish a direct connection.
- Technical Scope: Implementing an automated sync requires complex protocols that are beyond our current technical capacity.
- Current Workflow: Due to these constraints, we are currently limited to manual data collection (collecting files and importing them into the system).
- Operational Reality: This manual process is the only feasible way to move forward under our existing resources and access levels.
- Resource Constraints: Direct integration would require specialized IT support and infrastructure changes that are not available at this stage.

THE END

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Pham Minh Quang
范明光

Pegatron Corp